

Deep Learning Technical Report

1. Introduction

This report investigates the impact of different hyperparameters on the performance of a Convolutional Neural Network (CNN) trained on the CIFAR-100 dataset using PyTorch. The objective was to compare model performance across experiments and identify the most effective configuration.

2. Experimental Setup

- Framework: PyTorch
- Dataset: CIFAR-100 (100 classes, 50,000 training images, 10,000 test images)
- Model: SimpleCNN with two convolutional layers and two fully connected layers
- Baseline Parameters:
 - Learning Rate: 0.0001
 - Batch Size: 32
 - Epochs: 50
 - Loss Function: CrossEntropyLoss
 - Evaluation Metric: Accuracy (%)

3. Hyperparameters Tested

- Optimizers: SGD, Adam, RMSprop, Adagrad
- Batch Sizes: 16, 32, 64, 128
- Epoch Counts: 30, 50, 100

4. Results Summary

- Accuracy ranged between **44% and 49%**.
- Final loss values ranged between **1.7 and 2.9**.

Experiment Type Best Result		Observation
Optimizer	Adam (Accuracy 46.8%, Loss 2.7)	Outperformed SGD and RMSprop
Batch Size	64 (Accuracy 48.1%, Loss 1.7)	Larger batch improved stability

Experiment Type	Best Result	Observation
Epochs	100 (Accuracy 49.3%, Loss ~2.0)	Longer training improved accuracy

5. Visual Analysis

- Bar charts showed accuracy differences across experiments, highlighting Adam and larger batch sizes as superior.
- Scatter plots confirmed the inverse relationship between loss and accuracy — lower loss corresponded to higher accuracy.

6. Discussion Across experiments, the CNN model displayed stable performance, with accuracy consistently between 44% and 49%. Optimizer choice had the greatest impact, with Adam outperforming others due to its adaptive learning rate mechanism. Batch size also influenced results, with 64 providing the best balance between efficiency and accuracy. Longer training improved performance, but the marginal gains beyond 50 epochs suggest diminishing returns.

7. Conclusion The optimal configuration identified was:

- Optimizer: Adam
- Learning Rate: 0.0001
- Batch Size: 64
- Epochs: 100

This setup achieved the highest accuracy (49.3%) and lowest loss (~1.7). The experiments confirm that optimizer selection, batch size, and training duration are critical factors in CNN performance.